

Mathematical Foundations

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These notes develop the core mathematical toolkit for the course: the language of probability, three generating functions (MGF, CGF, characteristic function), Gaussian integration in arbitrary dimension, Wick's theorem, the saddle-point approximation, and the Hubbard–Stratonovich transformation. Every definition is followed by a worked example; every theorem is derived step by step.

Probability: basic definitions

A **sample space** Ω is the set of all possible outcomes of an experiment. A **random variable** X is a function $X : \Omega \rightarrow \mathbb{R}$ assigning a real number to each outcome. When X is continuous its distribution is described by a **probability density function** (pdf) $f_X(x)$ satisfying

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1. \quad (1)$$

The probability that X falls in $[a, b]$ is $\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$. The **cumulative distribution function** (CDF) is $F_X(x) = \int_{-\infty}^x f_X(t) dt$, so $f_X = F'_X$. When X is discrete we write $P(X = x_k) = p_k$ with $\sum_k p_k = 1$.

The **expectation** and **variance** are

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx, \quad \text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \quad (2)$$

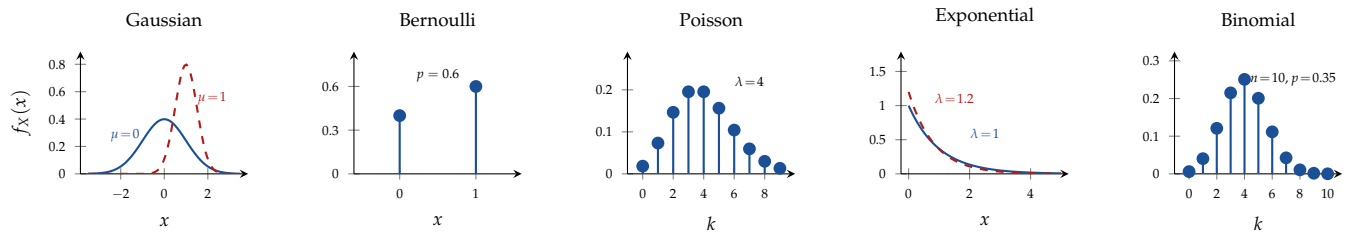


Figure 1: The five distributions introduced in this section. *Blue*: primary parameter choice; *dashed red*: alternative parameter.

Classic distributions

THE FIVE DISTRIBUTIONS shown in Figure ?? appear throughout the course.

Gaussian. $X \sim \mathcal{N}(\mu, \sigma^2)$ has pdf

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}. \quad (3)$$

It maximises entropy subject to fixed mean and variance, and is the attractor for sums of finite-variance variables (CLT).

Bernoulli. $X \sim \text{Bernoulli}(p)$: $P(X = x) = p^x(1-p)^{1-x}$, $x \in \{0, 1\}$. Mean p , variance $p(1-p)$.

Binomial. $X \sim \text{Binomial}(n, p)$: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. Mean np , variance $np(1-p)$.

Poisson. $X \sim \text{Poisson}(\lambda)$: $P(X = k) = e^{-\lambda} \lambda^k / k!$. Mean = variance = λ .

Exponential. $X \sim \text{Exp}(\lambda)$: $f_X(x) = \lambda e^{-\lambda x}$, $x \geq 0$. The only continuous memoryless distribution.

Generating functions

THREE GENERATING FUNCTIONS encode all statistical information in a distribution. They are the main calculational tools we will use to manipulate and identify distributions.

Moment generating function

The **moment generating function** (MGF) is

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx. \quad (4)$$

Expanding $e^{tX} = \sum_{n=0}^{\infty} (tX)^n / n!$ and taking expectations term by term,

$$M_X(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbb{E}[X^n], \quad (5)$$

so the n -th moment is recovered by differentiation:

$$\mathbb{E}[X^n] = \left. \frac{d^n M_X}{dt^n} \right|_{t=0}. \quad (6)$$

For independent X, Y , the MGF of the sum factorises: $M_{X+Y}(t) = M_X(t)M_Y(t)$, since $\mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX}]\mathbb{E}[e^{tY}]$ under independence.

MGF of the Gaussian

For $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx. \quad (7)$$

Combine the exponentials:

$$tx - \frac{(x - \mu)^2}{2\sigma^2} = -\frac{(x - (\mu + \sigma^2 t))^2}{2\sigma^2} + \mu t + \frac{\sigma^2 t^2}{2}. \quad (8)$$

The constant terms factor out; the remaining integral is a Gaussian normalisation equal to 1:

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right). \quad (9)$$

Differentiating at $t = 0$ gives $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$.

MGF of the Bernoulli

$M_X(t) = (1 - p) + pe^t$. Then $M'_X(0) = p$ and $\text{Var}(X) = M''_X(0) - (M'_X(0))^2 = p(1 - p)$.

Cumulant generating function

The **cumulant generating function** (CGF) is

$$K_X(t) = \log M_X(t), \quad (10)$$

and the n -th **cumulant** is $\kappa_n = K_X^{(n)}(0)$. The first two are $\kappa_1 = \mathbb{E}[X]$ and $\kappa_2 = \text{Var}(X)$; higher cumulants measure non-Gaussianity (κ_3 is skewness, κ_4 is excess kurtosis). All cumulants of a Gaussian beyond the second vanish.

WHY THE LOGARITHM? Because for independent X, Y ,

$$K_{X+Y}(t) = K_X(t) + K_Y(t). \quad (11)$$

Cumulants are additive for independent variables; moments are not. This makes the CGF the natural tool for large-system analysis.

CGF of the Gaussian

$K_X(t) = \mu t + \frac{1}{2}\sigma^2 t^2$. The CGF is a polynomial of degree 2 — the hallmark of the Gaussian. Hence $\kappa_1 = \mu$, $\kappa_2 = \sigma^2$, $\kappa_n = 0$ for $n \geq 3$.

CGF of the Poisson

The MGF is $M_X(t) = e^{-\lambda} \sum_k (\lambda e^t)^k / k! = \exp(\lambda(e^t - 1))$, so

$$K_X(t) = \lambda(e^t - 1), \quad \kappa_n = \lambda \quad \forall n \geq 1. \quad (12)$$

All cumulants equal λ , confirming $\mathbb{E}[X] = \text{Var}(X) = \lambda$.

Characteristic function

The **characteristic function** (Fourier transform of f_X) is

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} f_X(x) dx, \quad t \in \mathbb{R}. \quad (13)$$

It always exists since $|e^{itx}| = 1$, and it *uniquely* determines the distribution even when the MGF does not exist (heavy tails). The inversion formula reconstructs the pdf:

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_X(t) dt. \quad (14)$$

Gaussian. $\phi_X(t) = \exp(i\mu t - \frac{1}{2}\sigma^2 t^2)$, obtained by substituting $t \rightarrow it$ in (??).

Figure ?? summarises the relationships among the three generating functions.

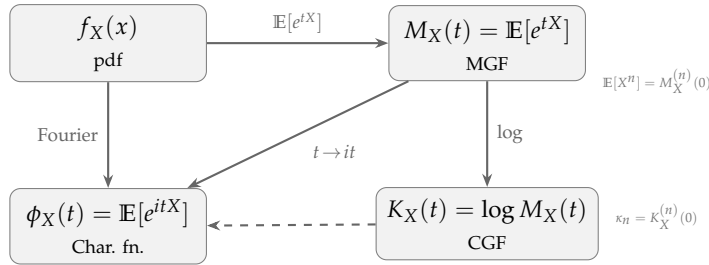


Figure 2: Relationships among the three generating functions. Arrows show how each is constructed from the previous.

Multivariate distributions

FOR TWO RANDOM VARIABLES X, Y the **joint pdf** $f_{X,Y}(x, y)$ satisfies $\iint f_{X,Y} dx dy = 1$. The **marginal** pdfs are

$$f_X(x) = \int f_{X,Y}(x, y) dy, \quad f_Y(y) = \int f_{X,Y}(x, y) dx. \quad (15)$$

The **conditional** pdf is $f_{X|Y}(x|y) = f_{X,Y}(x, y) / f_Y(y)$, which is Bayes' theorem in density form: $f_{X|Y}(x|y) \propto f_{Y|X}(y|x) f_X(x)$.

X and Y are **independent** iff $f_{X,Y} = f_X f_Y$. The **covariance** and **correlation** are

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \quad \rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \in [-1, 1]. \quad (16)$$

The multivariate Gaussian

A random vector $\mathbf{x} \in \mathbb{R}^N$ follows $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$ if

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^N \det \mathbf{C}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{C}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right). \quad (17)$$

We derive this form from first principles in Section ??.

Gaussian integrals

GAUSSIAN INTEGRALS are the fundamental computational engine of statistical physics and machine learning theory. The ability to evaluate them in closed form — even in hundreds of dimensions — is what makes Gaussian models analytically tractable.

The one-dimensional integral

$$\int_{-\infty}^{\infty} e^{-\frac{1}{2}ax^2} dx = \sqrt{\frac{2\pi}{a}}, \quad a > 0. \quad (18)$$

Derivation. Let $I = \int e^{-\frac{a}{2}x^2} dx$ and square it:

$$I^2 = \iint e^{-\frac{a}{2}(x^2+y^2)} dx dy. \quad (19)$$

Switch to polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, with area element $r dr d\theta$:

$$I^2 = 2\pi \int_0^{\infty} r e^{-\frac{a}{2}r^2} dr. \quad (20)$$

Substitute $u = \frac{a}{2}r^2$ so $r dr = du/a$:

$$I^2 = \frac{2\pi}{a} \int_0^{\infty} e^{-u} du = \frac{2\pi}{a}, \quad (21)$$

giving $I = \sqrt{2\pi/a}$. The polar-coordinates trick is illustrated in the margin.

Adding a linear source term by completing the square, $-\frac{a}{2}x^2 + bx = -\frac{a}{2}(x - b/a)^2 + b^2/(2a)$, and shifting $x \rightarrow x + b/a$:

$$\int_{-\infty}^{\infty} e^{-\frac{1}{2}ax^2+bx} dx = \sqrt{\frac{2\pi}{a}} e^{\frac{b^2}{2a}}. \quad (22)$$

The multidimensional integral

Let $\mathbf{x} \in \mathbb{R}^N$, $\mathbf{A}$ real symmetric positive-definite, $\mathbf{b} \in \mathbb{R}^N$. Then:

$$\int_{\mathbb{R}^N} d^N x \exp\left(-\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x}\right) = \sqrt{\frac{(2\pi)^N}{\det \mathbf{A}}} \exp\left(\frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}\right). \quad (23)$$

Step 1: Complete the square. Claim:

$$-\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} = -\frac{1}{2}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b})^T \mathbf{A}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b}) + \frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}. \quad (24)$$

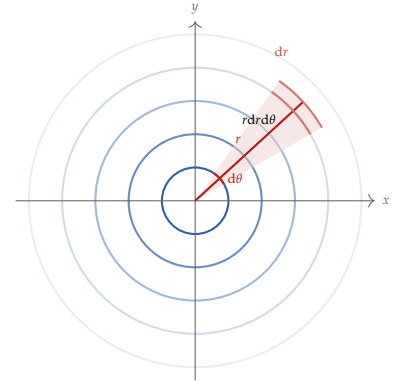


Figure 3: Squaring I converts the product of two 1-D Gaussians into a 2-D radially symmetric integral. The area element in polar coordinates is $r dr d\theta$.

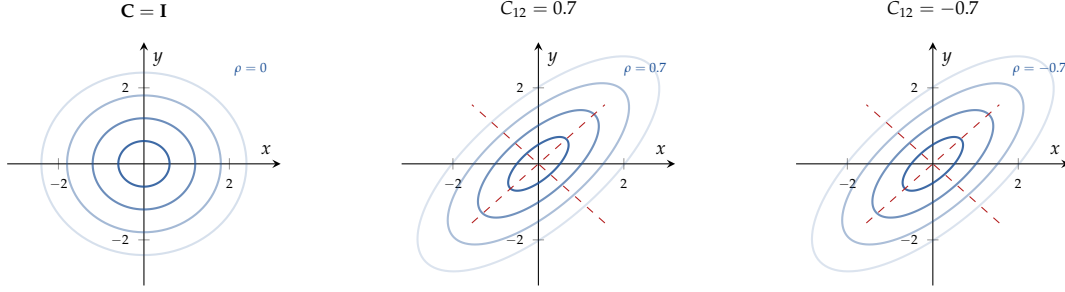


Figure 4: Level sets (constant-density contours) of 2-D Gaussians. Left: isotropic ($C = I$), circular contours. Centre: positive correlation $\rho = 0.7$, ellipse tilted at 45° . Right: negative correlation $\rho = -0.7$, ellipse tilted at -45° . Dashed red lines mark the principal axes (eigenvectors of C).

Expanding the right-hand side and using $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ and symmetry of $\mathbf{A}$:

$$\begin{aligned} \text{RHS} &= -\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \frac{1}{2}\mathbf{x}^T\mathbf{b} + \frac{1}{2}\mathbf{b}^T\mathbf{x} \\ &\quad - \frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b} + \frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b} \\ &= -\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x}, \end{aligned} \quad (25)$$

confirming (??).

Step 2: Shift the variable. Set $\mathbf{y} = \mathbf{x} - \mathbf{A}^{-1}\mathbf{b}$ ($d^N y = d^N x$):

$$\int d^N x e^{-\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x}} = e^{\frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}} \int d^N y e^{-\frac{1}{2}\mathbf{y}^T\mathbf{A}\mathbf{y}}. \quad (26)$$

Step 3: Diagonalise $\mathbf{A}$. By the spectral theorem $\mathbf{A} = \mathbf{O}^T\mathbf{\Lambda}\mathbf{O}$ with $\mathbf{O}$ orthogonal and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_N)$, $\lambda_i > 0$. Set $\mathbf{u} = \mathbf{O}\mathbf{y}$, so $d^N u = d^N y$ and $\mathbf{y}^T\mathbf{A}\mathbf{y} = \sum_i \lambda_i u_i^2$. The integral factorises:

$$\int d^N y e^{-\frac{1}{2}\mathbf{y}^T\mathbf{A}\mathbf{y}} = \prod_{i=1}^N \sqrt{\frac{2\pi}{\lambda_i}} = \sqrt{\frac{(2\pi)^N}{\det \mathbf{A}}}. \quad (27)$$

Combining (??) and (??) gives (??).

The source trick and moment generation

Define the **generating functional**¹

$$Z(\mathbf{b}) = \int d^N x e^{-\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x}} = Z_0 e^{\frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}}, \quad (28)$$

where $Z_0 = Z(0) = \sqrt{(2\pi)^N / \det \mathbf{A}}$. Since $\partial e^{\mathbf{b}^T\mathbf{x}} / \partial b_i = x_i e^{\mathbf{b}^T\mathbf{x}}$, differentiating and setting $\mathbf{b} = \mathbf{0}$ extracts moments:

$$\langle x_i \rangle = \frac{1}{Z_0} \frac{\partial Z}{\partial b_i} \Big|_{\mathbf{b}=\mathbf{0}}. \quad (29)$$

First moment. $\partial Z / \partial b_i = Z(\mathbf{b})(\mathbf{A}^{-1}\mathbf{b})_i$. At $\mathbf{b} = \mathbf{0}$: $\langle x_i \rangle = 0$.

Second moment. Differentiating again:

$$\frac{\partial^2 Z}{\partial b_i \partial b_j} \Big|_{\mathbf{b}=\mathbf{0}} = Z_0 (\mathbf{A}^{-1})_{ij}, \quad (30)$$

¹ The name *generating functional* comes from field theory; here it is simply the Gaussian integral as a function of the source $\mathbf{b}$.

so

$$\langle x_i x_j \rangle = (\mathbf{A}^{-1})_{ij} = C_{ij}. \quad (31)$$

The two-point correlator equals the (i, j) entry of the inverse of the matrix in the exponent. We call C_{ij} the **propagator**. The level sets of 2-D Gaussians for different covariance structures are shown in Figure ??.

Wick's theorem

WICK'S THEOREM gives an algorithmic rule for computing any moment of a Gaussian in terms of the propagator alone. It is the central combinatorial identity of perturbative field theory.

Statement

Let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$. Then:

- (1) All odd moments vanish.
- (2) All even moments satisfy:

$$\langle x_{i_1} x_{i_2} \cdots x_{i_{2n}} \rangle = \sum_{\text{pairings}} \prod_{\text{pairs } (j,k)} C_{i_j i_k}, \quad (32)$$

where the sum runs over all $(2n - 1)!! = 1 \cdot 3 \cdot 5 \cdots (2n - 1)$ distinct ways to pair $\{i_1, \dots, i_{2n}\}$ into n unordered pairs.

Derivation via the generating functional

With $Z(\mathbf{b}) = Z_0 e^{\frac{1}{2} \mathbf{b}^T \mathbf{C} \mathbf{b}}$, expand:

$$e^{\frac{1}{2} \mathbf{b}^T \mathbf{C} \mathbf{b}} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{2} \sum_{i,j} b_i C_{ij} b_j \right)^n. \quad (33)$$

The m -th moment is $\langle x_{i_1} \cdots x_{i_m} \rangle = \partial^m e^{\frac{1}{2} \mathbf{b}^T \mathbf{C} \mathbf{b}} / \partial b_{i_1} \cdots \partial b_{i_m} |_{\mathbf{b}=\mathbf{0}}$.

For odd $m = 2n + 1$: every term in (??) has even powers of $\mathbf{b}$; after $2n + 1$ derivatives at least one b factor remains, which vanishes at $\mathbf{0}$. All odd moments vanish.

For even $m = 2n$: only the n -th term in (??) survives. Expanding it,

$$\frac{1}{n! 2^n} \sum_{i_1, j_1} \cdots \sum_{i_n, j_n} C_{i_1 j_1} \cdots C_{i_n j_n} b_{i_1} b_{j_1} \cdots b_{i_n} b_{j_n}. \quad (34)$$

Differentiating $2n$ times matches each b_{k_l} to exactly one index; the $\frac{1}{n! 2^n}$ prefactor precisely cancels the $n! 2^n$ over-counting of pairings, leaving the sum over distinct pairings of products of $C_{k_a k_b}$. This is Eq. (??).

Worked examples

Fourth moment

For $X \sim \mathcal{N}(0, \sigma^2)$ there are $(4 - 1)!! = 3$ pairings of $\{1, 2, 3, 4\}$:

- $(1, 2)(3, 4)$: contributes $\sigma^2 \cdot \sigma^2 = \sigma^4$.
- $(1, 3)(2, 4)$: contributes σ^4 .
- $(1, 4)(2, 3)$: contributes σ^4 .

So $\langle X^4 \rangle = 3\sigma^4$. (The three pairings are illustrated in the margin.)

Verification via the MGF: differentiating $e^{\frac{1}{2}\sigma^2 t^2}$ four times at $t = 0$ gives $3\sigma^4$. ✓

Two-variable moment

For $(x_1, x_2) \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$, the three pairings of $(1, 1, 2, 2)$ give

$$\langle x_1^2 x_2^2 \rangle = C_{11}C_{22} + 2C_{12}^2. \quad (35)$$

For independent variables ($C_{12} = 0$) this reduces to $\langle x_1^2 \rangle \langle x_2^2 \rangle$. ✓

Sixth moment

$(6 - 1)!! = 15$ pairings, each contributing σ^6 , so $\langle X^6 \rangle = 15\sigma^6$.

Saddle-point approximation

MANY INTEGRALS in statistical physics and machine learning take the form

$$I(N) = \int_{-\infty}^{\infty} dx e^{-Nf(x)}, \quad (36)$$

where N is a large parameter (number of neurons, spins, data points) and $f(x)$ is smooth. As $N \rightarrow \infty$, the integrand concentrates exponentially near the minimum of f .

One-dimensional case

Let x^* minimise f : $f'(x^*) = 0$, $f''(x^*) > 0$. Taylor expand:

$$f(x) = f(x^*) + \underbrace{f'(x^*)}_{0}(x - x^*) + \frac{f''(x^*)}{2}(x - x^*)^2 + \dots \quad (37)$$

Substituting and evaluating the Gaussian integral:

$$I(N) \approx \sqrt{\frac{2\pi}{Nf''(x^*)}} e^{-Nf(x^*)}, \quad N \rightarrow \infty. \quad (38)$$

The cubic correction is $\mathcal{O}(N^{-1/2}) \rightarrow 0$ since fluctuations are of order $(x - x^*) \sim N^{-1/2}$.

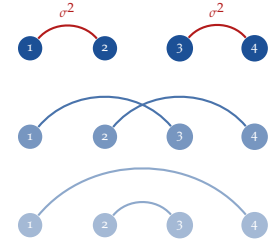


Figure 5: The three Wick pairings of $\{1, 2, 3, 4\}$ for computing $\langle X^4 \rangle$. Each arc contributes a propagator σ^2 .

Saddle-point concentration

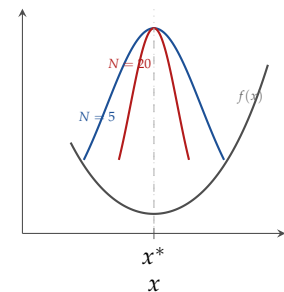


Figure 6: As N grows, $e^{-Nf(x)}$ concentrates ever more sharply near x^* . The saddle-point approximation replaces f with its quadratic expansion around x^* .

Multidimensional case

For $\mathbf{x}^*$ with $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and Hessian $\mathbf{H}_{ij} = \partial^2 f / \partial x_i \partial x_j|_{\mathbf{x}^*}$:

$$\int_{\mathbb{R}^N} d^N x e^{-Nf(\mathbf{x})} \approx \left(\frac{2\pi}{N} \right)^{N/2} \frac{e^{-Nf(\mathbf{x}^*)}}{\sqrt{\det \mathbf{H}(\mathbf{x}^*)}}. \quad (39)$$

Example: Stirling's approximation

From $n! = \int_0^\infty x^n e^{-x} dx$, substitute $x = nu$:

$$n! = n^{n+1} \int_0^\infty e^{-n(u - \log u)} du. \quad (40)$$

Identify $f(u) = u - \log u$. Saddle: $f'(1) = 0$, $f''(1) = 1$, $f(1) = 1$. Applying (??):

$$\log n! \approx n \log n - n + \frac{1}{2} \log(2\pi n). \quad (41)$$

Example: Entropy of a spin system

The number of configurations of N spins with magnetisation $m = \frac{1}{N} \sum_i \sigma_i$ is $\Omega(m) = \binom{N}{N(1+m)/2}$. Using Stirling with $n_\pm = N(1 \pm m)/2$:

$$s(m) \equiv \frac{\log \Omega}{N} = -\frac{1+m}{2} \log \frac{1+m}{2} - \frac{1-m}{2} \log \frac{1-m}{2}. \quad (42)$$

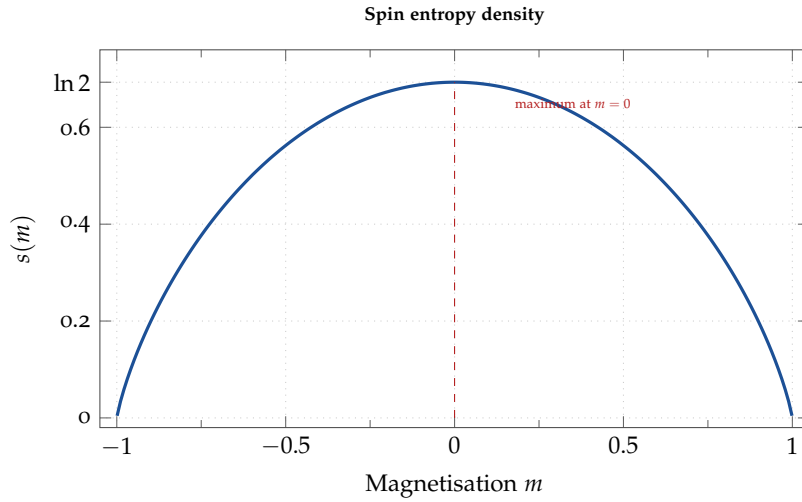


Figure 7: The spin entropy density $s(m)$ from Eq. (42). It is maximised at $m = 0$ (maximum disorder, $s = \log 2$) and vanishes at $m = \pm 1$ (all spins aligned).

Hubbard–Stratonovich transformation

THE HUBBARD–STRATONOVICH (HS) TRANSFORMATION is the central algebraic trick of mean-field theory. It converts an intractable quadratic coupling between many degrees of freedom into an integral over a single auxiliary field, making the saddle-point approximation applicable.

The basic identity

$$e^{\frac{a^2}{2}} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-\frac{z^2}{2} + az}. \quad (43)$$

Proof. Complete the square in z : $-\frac{z^2}{2} + az = -\frac{1}{2}(z - a)^2 + \frac{a^2}{2}$. Then:

$$\frac{1}{\sqrt{2\pi}} \int e^{-\frac{z^2}{2} + az} dz = \frac{e^{a^2/2}}{\sqrt{2\pi}} \int e^{-\frac{(z-a)^2}{2}} dz = \frac{e^{a^2/2}}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = e^{a^2/2}. \quad \checkmark \quad (44)$$

General rescaled form

For $\alpha > 0$ and coupling J , substituting $a = J/\sqrt{\alpha}$ and $z \rightarrow \sqrt{\alpha}w$ in (43):

$$e^{\frac{J^2}{2\alpha}} = \sqrt{\frac{\alpha}{2\pi}} \int_{-\infty}^{\infty} dz e^{-\frac{\alpha}{2}z^2 + Jz}. \quad (45)$$

Derivation. Start from (43) with $a = J/\sqrt{\alpha}$. Substitute $z = \sqrt{\alpha}w$:

$$e^{\frac{J^2}{2\alpha}} = \frac{1}{\sqrt{2\pi}} \int e^{-\frac{z^2}{2} + \frac{Jz}{\sqrt{\alpha}}} dz = \frac{\sqrt{\alpha}}{\sqrt{2\pi}} \int e^{-\frac{\alpha w^2}{2} + Jw} dw. \quad (46)$$

Decoupling pairwise interactions

The key application: a Boltzmann weight $e^{\frac{\beta J}{2N}(\sum_i \sigma_i)^2}$ couples every pair of spins. Setting $\alpha = N\beta J$ and $J_{\text{eff}} = \beta J \sum_i \sigma_i$ in (45):

$$e^{\frac{\beta J}{2N}(\sum_i \sigma_i)^2} = \sqrt{\frac{N\beta J}{2\pi}} \int dm \exp\left(-\frac{N\beta J}{2}m^2 + \beta Jm \sum_{i=1}^N \sigma_i\right). \quad (47)$$

After this transformation the exponent is *linear* in each σ_i : the spins see only the uniform external field βJm and are completely decoupled from one another.

Application: Curie–Weiss mean-field theory

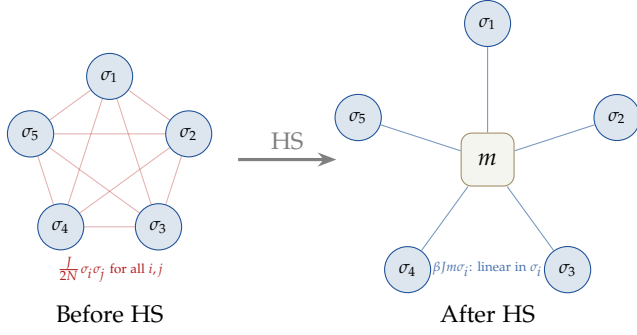


Figure 8: The Hubbard-Stratonovich transformation. Left: the original problem has all-to-all quadratic interactions $J\sigma_i\sigma_j/2N$, making the partition function a sum of 2^N entangled terms. Right: after introducing the auxiliary field m , each spin interacts only with m through a linear coupling $\beta J m \sigma_i$; the spins decouple and the sum factorises.

THE CURIE-WEISS MODEL is the fully-connected Ising model:

$$H = -\frac{J}{2N} \sum_{i,j} \sigma_i \sigma_j = -\frac{J}{2N} \left(\sum_i \sigma_i \right)^2, \quad \sigma_i \in \{+1, -1\}. \quad (48)$$

The partition function is $Z = \sum_{\{\sigma_i\}} e^{-\beta H}$.

Step 1 — Apply HS.

$$Z = \sqrt{\frac{N\beta J}{2\pi}} \int dm e^{-\frac{N\beta J}{2} m^2} \sum_{\{\sigma_i\}} \prod_{i=1}^N e^{\beta J m \sigma_i}. \quad (49)$$

Step 2 — Factorise. Spins are independent after HS:

$$\sum_{\{\sigma_i\}} \prod_i e^{\beta J m \sigma_i} = \prod_i (e^{\beta J m} + e^{-\beta J m}) = (2 \cosh \beta J m)^N. \quad (50)$$

Step 3 — Write as single integral.

$$Z = \sqrt{\frac{N\beta J}{2\pi}} \int dm \exp(-Nf(m)), \quad f(m) = \frac{\beta J}{2} m^2 - \log(2 \cosh \beta J m). \quad (51)$$

Step 4 — Saddle-point. $f'(m^*) = \beta J [m^* - \tanh(\beta J m^*)] = 0$:

$$m^* = \tanh(\beta J m^*). \quad (52)$$

This is the **mean-field self-consistency equation**. The auxiliary field m is the magnetisation $m^* = \langle \sigma_i \rangle$.

Step 5 — Phase structure. Non-trivial solutions $m^* \neq 0$ require the slope of $\tanh(\beta J m)$ at $m = 0$ to exceed 1:

$$\beta J > 1 \iff T < T_c \equiv J. \quad (53)$$

Above T_c only the paramagnetic solution $m^* = 0$ exists. Below T_c two ferromagnetic solutions $m^* = \pm m_0$ appear — this is the **ferromagnetic phase transition**, illustrated in Figure ?? . The entire derivation (HS $\rightarrow$ spin factorisation $\rightarrow$ saddle-point $\rightarrow$ self-consistency equation) is the prototype for every mean-field calculation in this course, including the Hopfield model and the replica method.

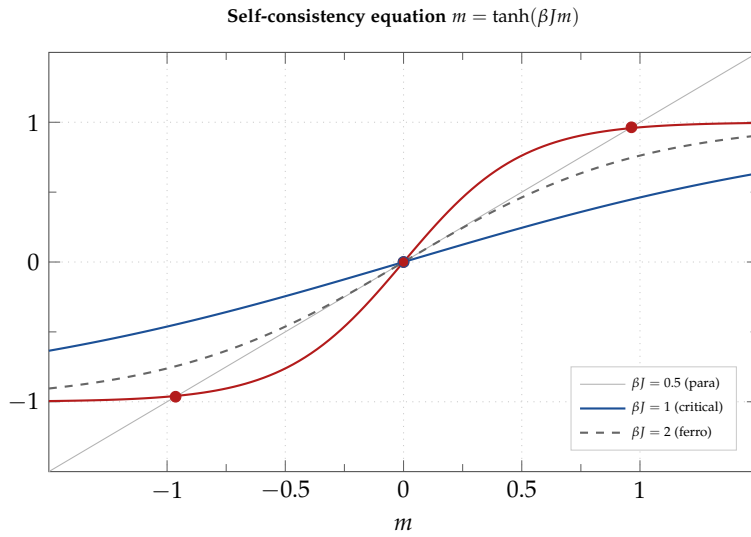


Figure 9: Graphical solution of the self-consistency equation (??). Intersections of $y = \tanh(\beta J m)$ with the diagonal $y = m$ give the solutions m^* . Below T_c ($\beta J > 1$, red curve) three intersections appear: $m^* = 0$ (unstable) and $m^* = \pm m_0$ (stable, filled dots). Above T_c ($\beta J < 1$, blue) only $m^* = 0$ exists.